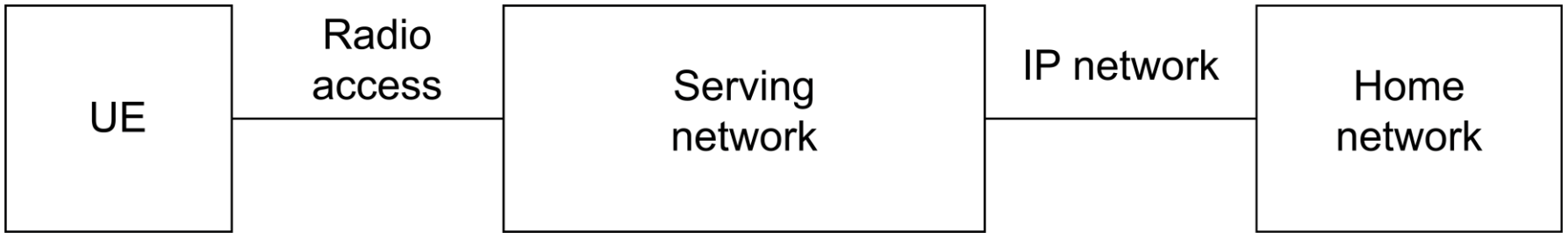
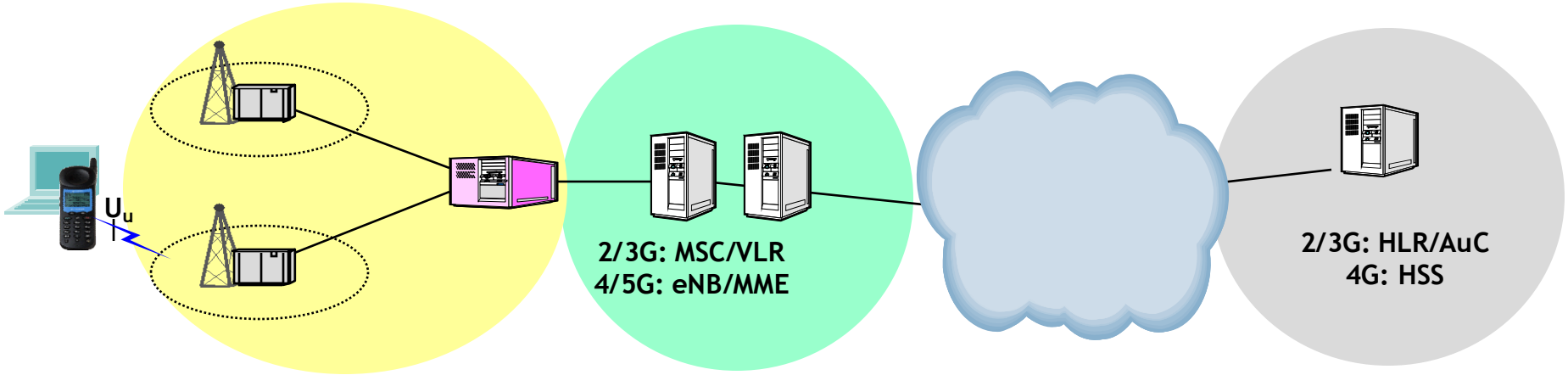
Security in Wireless Cellular systems

(focus limited to authentication, at this stage)

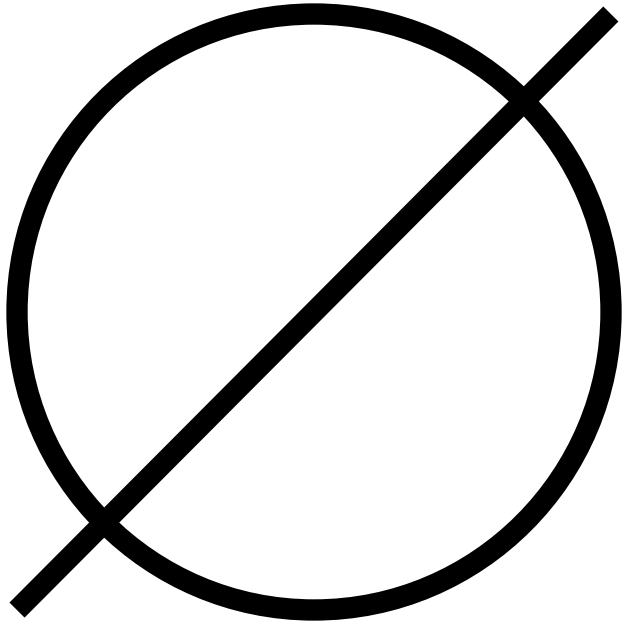
Architecture of wireless cellular systems (abstracted)





Security in 1G





Security in 2G

Security? No problem!
Let me do it!!



Can you
Protect us?



COMP-128 Security by obscurity

No mutual auth (Rogue BS)

No core network security



Security in 3G





(fairly) good ciphers - public scrutiny!

Encryption AND (in part) Integrity

Multual authentication

Core network security

Multiple keys

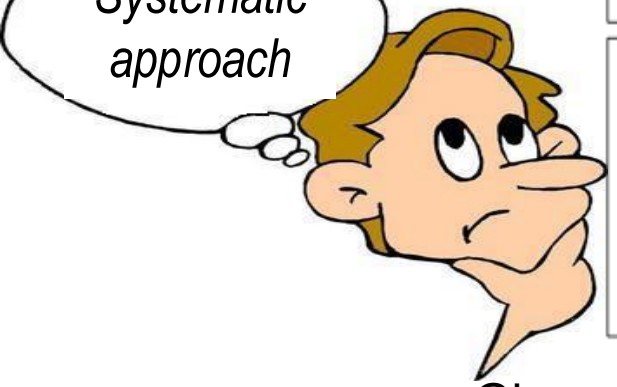
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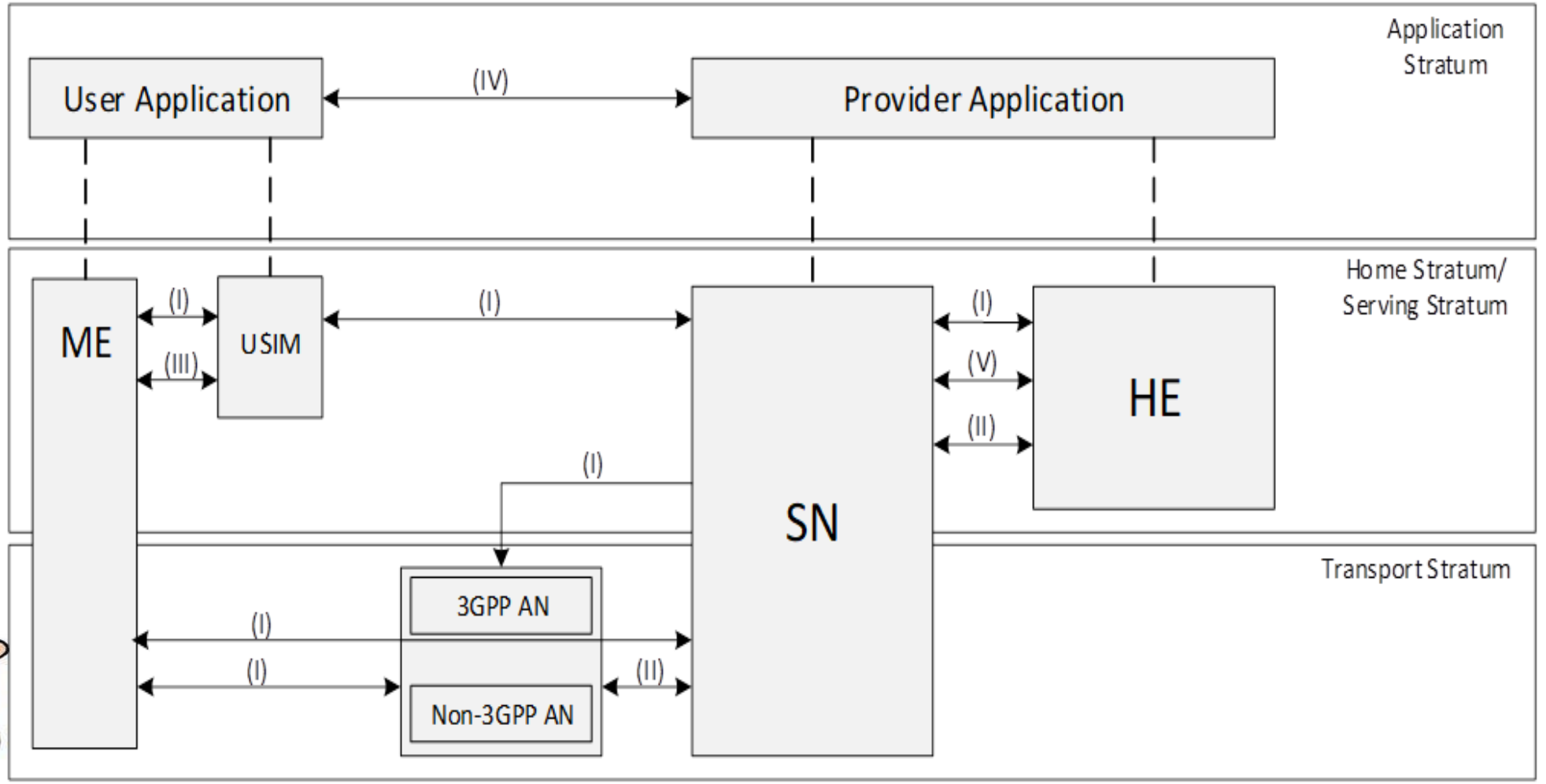
Security in 4G

Time for a Security architecture!



Systematic approach





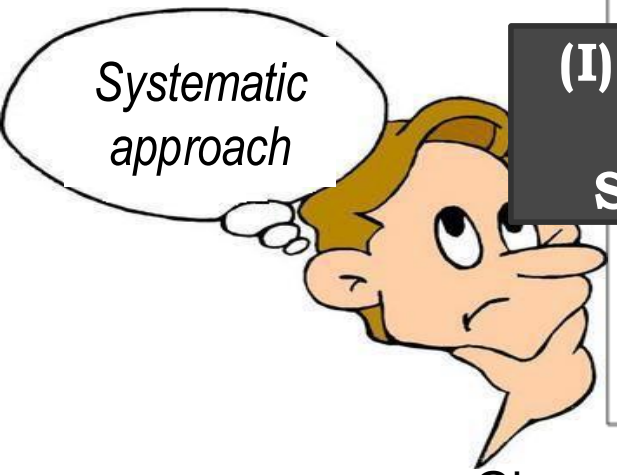
Security in 4G and 5G

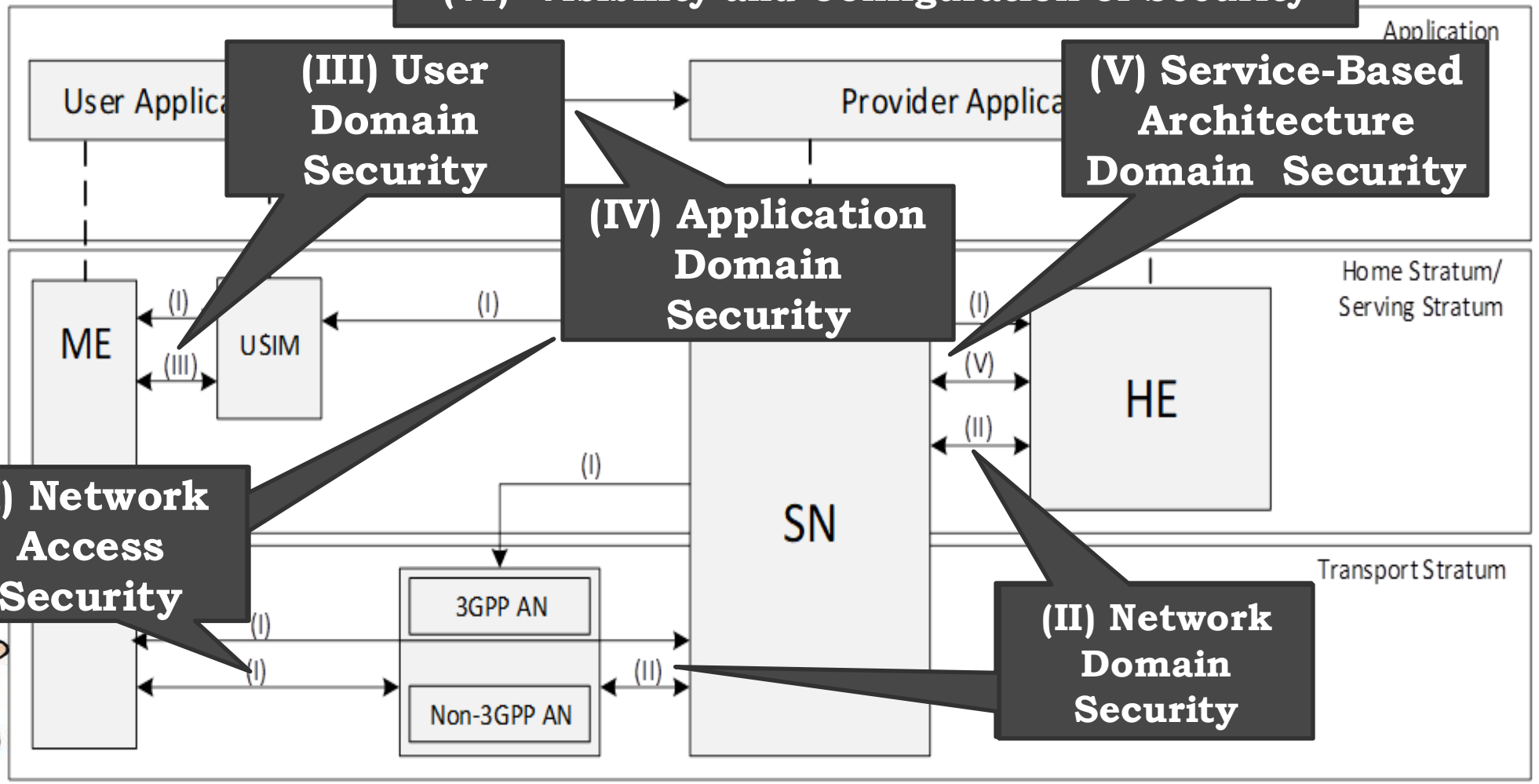
Actually… this is already the 5G security architecture… to save one slide (couple of differences over 4G)!

Time for a Security architecture!



Systematic approach





5G Security Architecture -

The 5G «vault» ☺: User data (UDR) and credentials (ARPF) used by UDM authentication/identification functions

Manage Network Access Security
(AMF+SEAF co-implemented)

No security support in DU (deployed in untrusted sites)

Test Framework SCAS for Security Assurance



non3GPP access

N3IWF

DU

CU

gNB

SEPP

AUSF

SIDF

ARPF

UDM

UDR

Security for non-3GPP access (Wi-Fi, etc)
key for later reuse

Improved AUTH and home control via Proof-of-Presence

Access Stratum Security terminated at CU

Security for interconnection traffic

omains)

G/6G network!

geo-political security debate is in part ill-posed)

L2 message integrity?

Breaking LTE on Layer Two

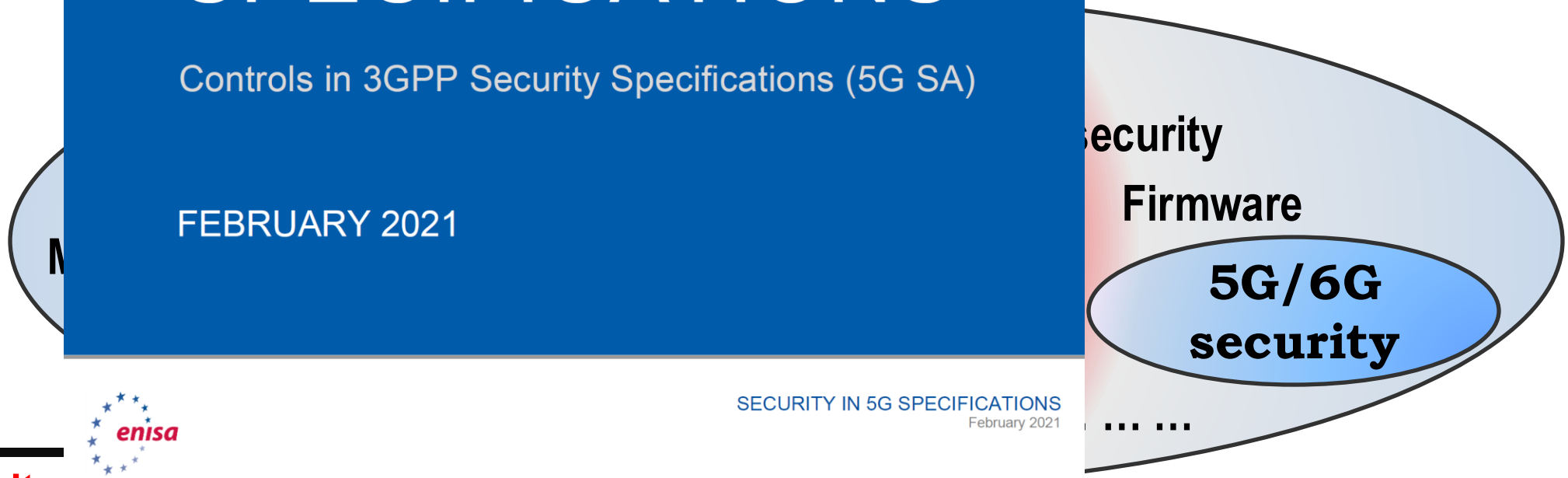
David Rupprecht, Ruhr-University Bochum, david.rupprecht@rub.de
Katharina Kohls, Ruhr-University Bochum, katharina.kohls@rub.de
Thorsten Holz, Ruhr-University Bochum, thorsten.holz@rub.de
Christina Pöpper, New York University Abu Dhabi, christina.poepper@nyu.edu

DU
CU
N3IWF
Protection Proxy

ARPF Auth credential Repository & Processing Fct
UDM Unified Data Management
UDR Unified Data Repository

But sys[illegible]nplex

5G/6G systems' security: A small part of a bigger picture!!



SECURITY IN 5G SPECIFICATIONS

Controls in 3GPP Security Specifications (5G SA)

FEBRUARY 2021

SECURITY IN 5G SPECIFICATIONS
February 2021

enisa

No 5G security "
SW/HW implementation
is insecure!

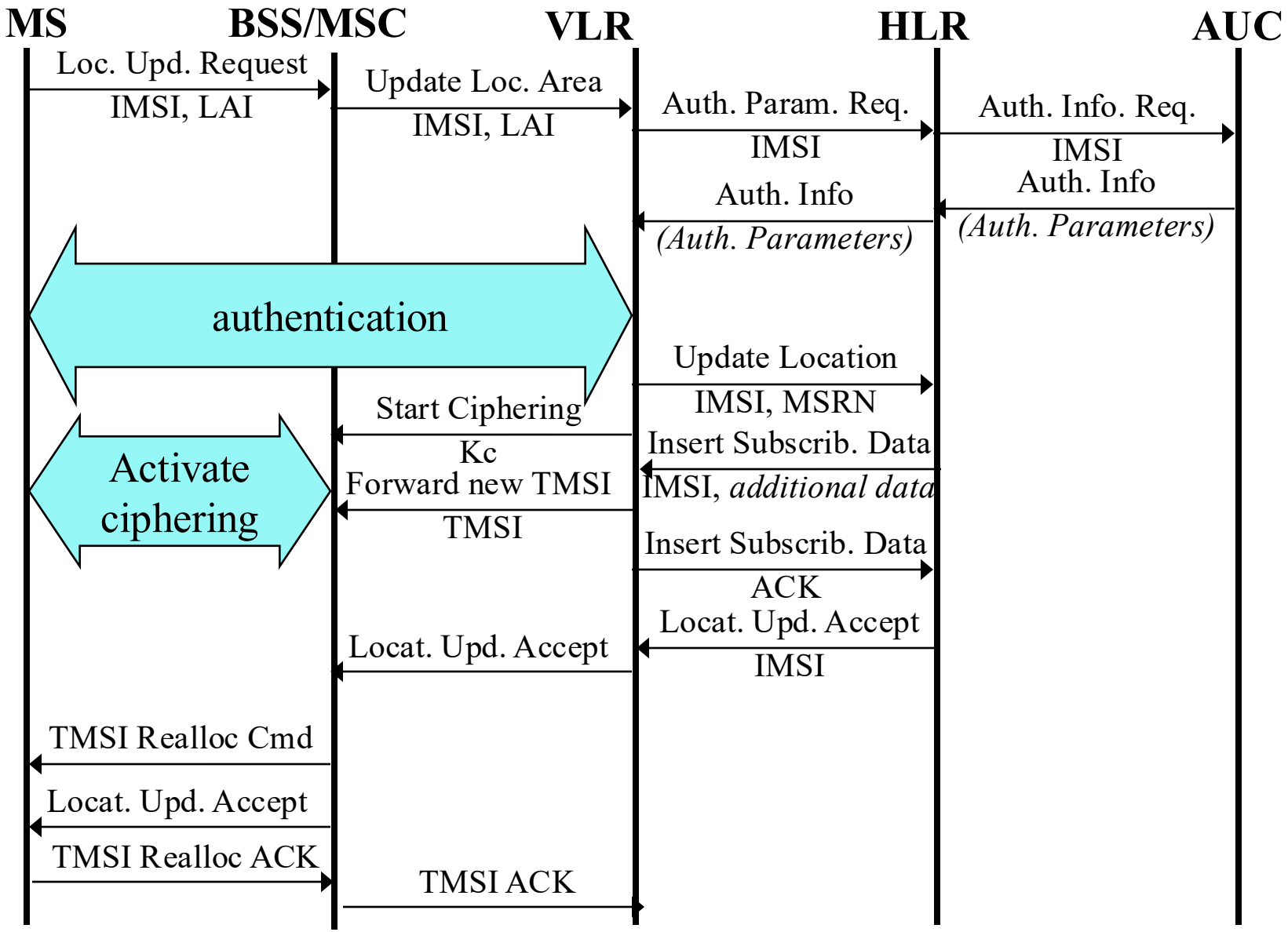
Critical infrastructure = high tier threats
«They» might ***design*** vulnerabilities!

SW networks, third party suppliers:
Complexity, increased threat surface!

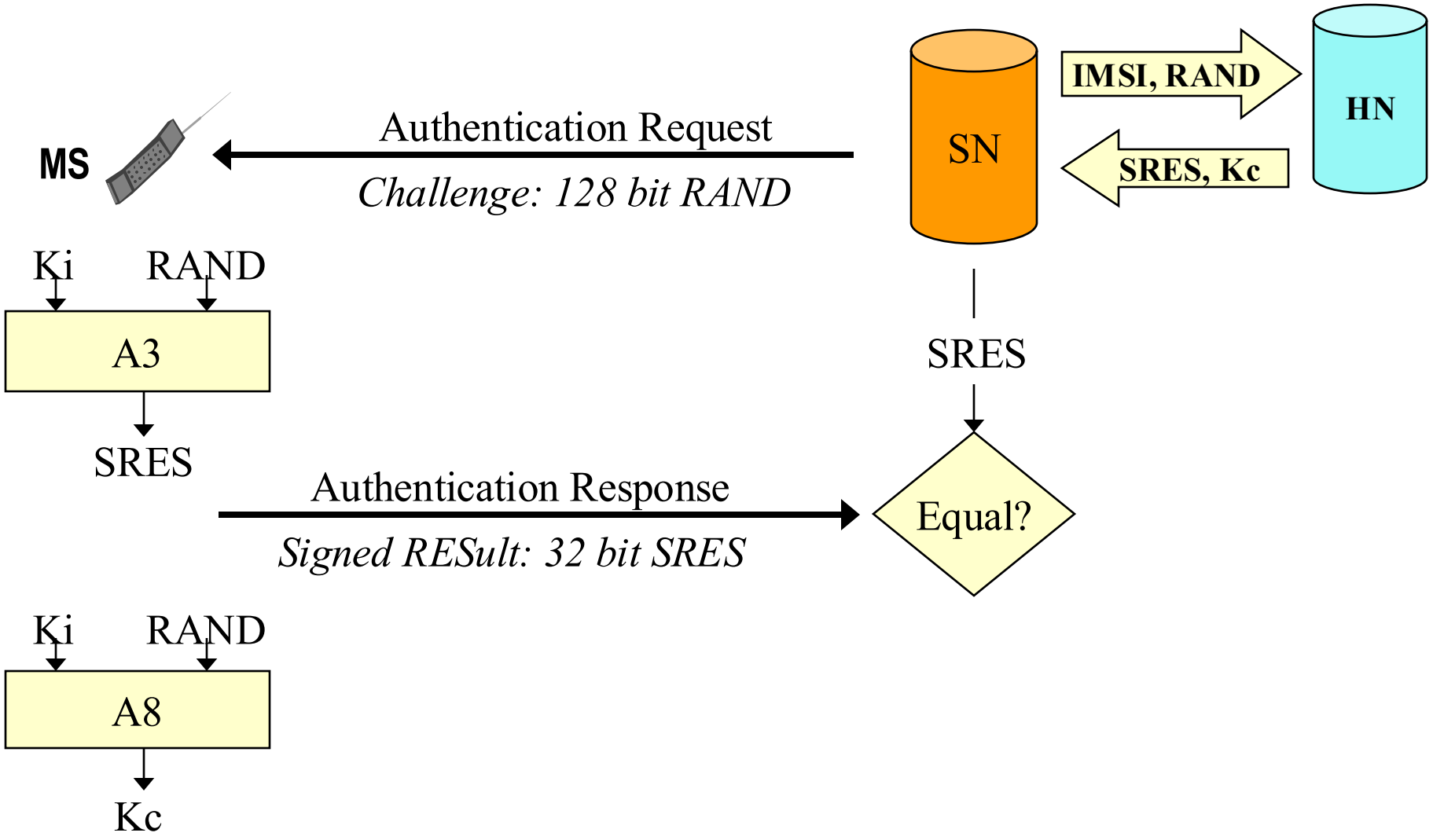
Too many «optional/should»
In the standard!
Very risky! (you might believe you are secured, while…)

Challenge-Response authentication in Wireless Cellular systems

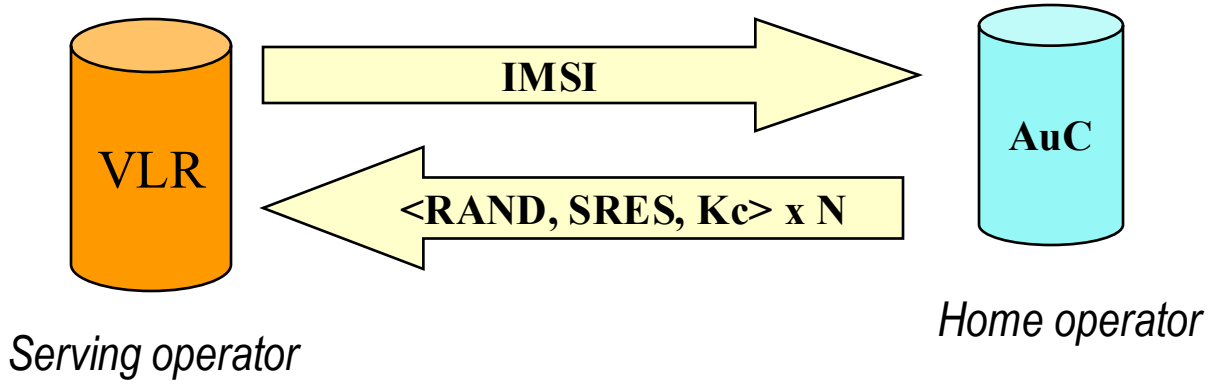
Authentication: when (e.g. 2/3G)



Authentication in 2G
(unilateral)



Triplets (Authentication Vector)



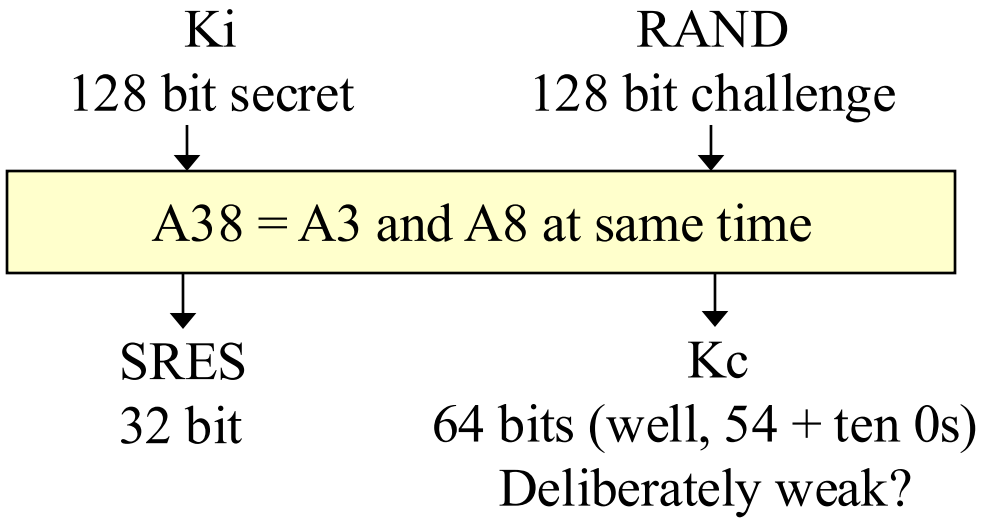
➔ **Idea: once in a VLR area, autentication will need to be performed MANY times**

➔ **Hence deliver N triplets, to be used for N distinct authentications**

➔ **IMPORTANT: VLR does NOT need to know authentication algo used (A3, A8)**

- ⇨ *Triplet contains computed result by AuC*
- ⇨ *A3, A8 run inside the SIM (given by operator)*

Authentication: details



➔Challenge response with:

- ⇨ *Challenge* → *RAND*
- ⇨ *Secret* → *Ki*
- ⇨ *Hash* → *A3 algorithm*

On the A3/A8 algorithms

- **Security by obscurity**
  - *A3 algorithm CAN BE operator-specific!*
  - *But most vendors originally used algo today called COMP128*
  - *Non disclosed but…*
    - Reverse engineered? Leaked out?
- **COMP128 broken in 1998**
    - http://www.isaac.cs.berkeley.edu/isaac/gsm-faq.html
  - *Chosen challenge attack [1998, Briceno, Goldberg, Wagner]*
    - Ki disclosed through about 150.000 queries with suitably selected challenges → approx 8h in 1998
    - Having Ki means cloning the card!
  - *Better attack [2002, Rao, Rohatgi, Scherzer, Tinguely]*
    - Less than 1 minute!
- **Lesson learned:**
  - *Security by obscurity does NOT work!! Leave hash design to crypto experts*

Over the air attach

➔**No mutual authentication!**

➔**Rogue BTS may easily perform the attach**

⇨*run it for sufficient time*

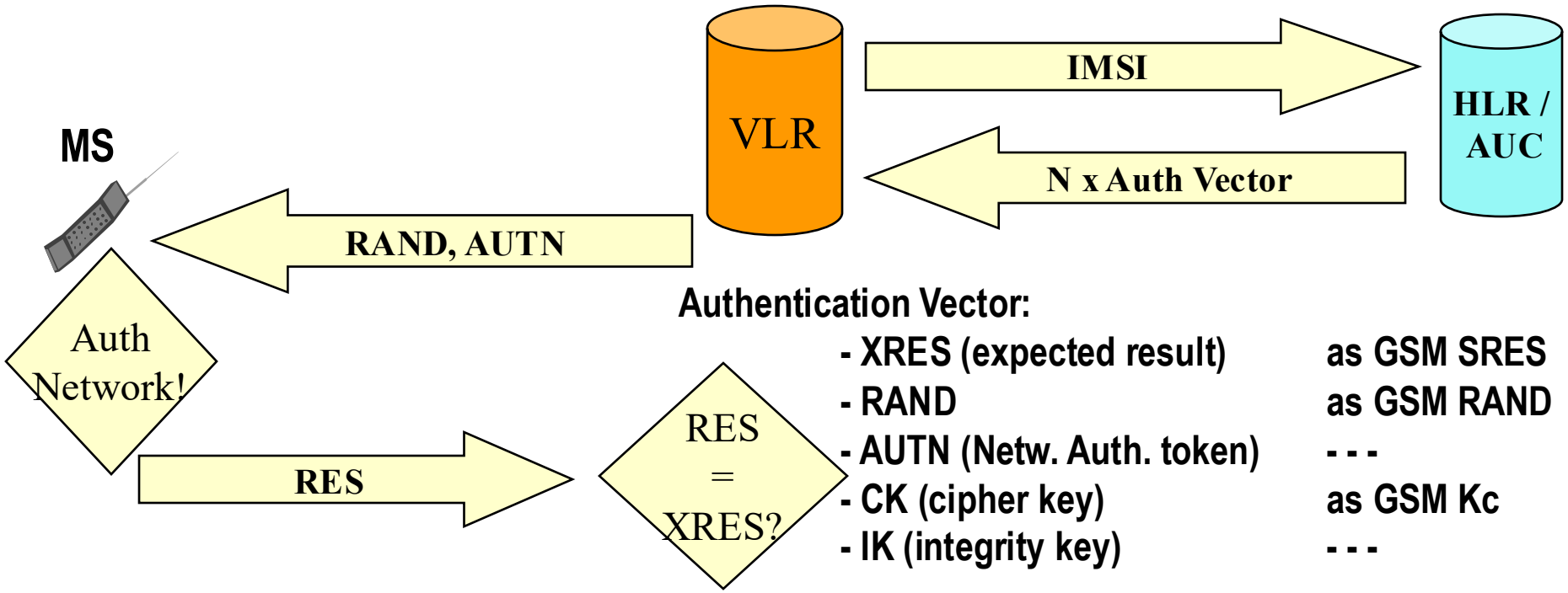
3G/4G/5G authentication: AKA Authentication and Key Agreement

(STRONGLY Simplified for our purposes; 3G terminology)

Major differences with 2G

- **➔Mutual authentication!**
  - ⇨*Optimized…*
  - ⇨*Guaranteed freshness for auth parameters*
- **➔More comprehensive security**
  - ⇨*More keys and several extra details*
    - →(we will not focus on this)
- **➔Algorithms FIRST scrutinized by research community, THEN selected**
  - ⇨*The opposite of security by obscurity* ☺

Authentication Vector

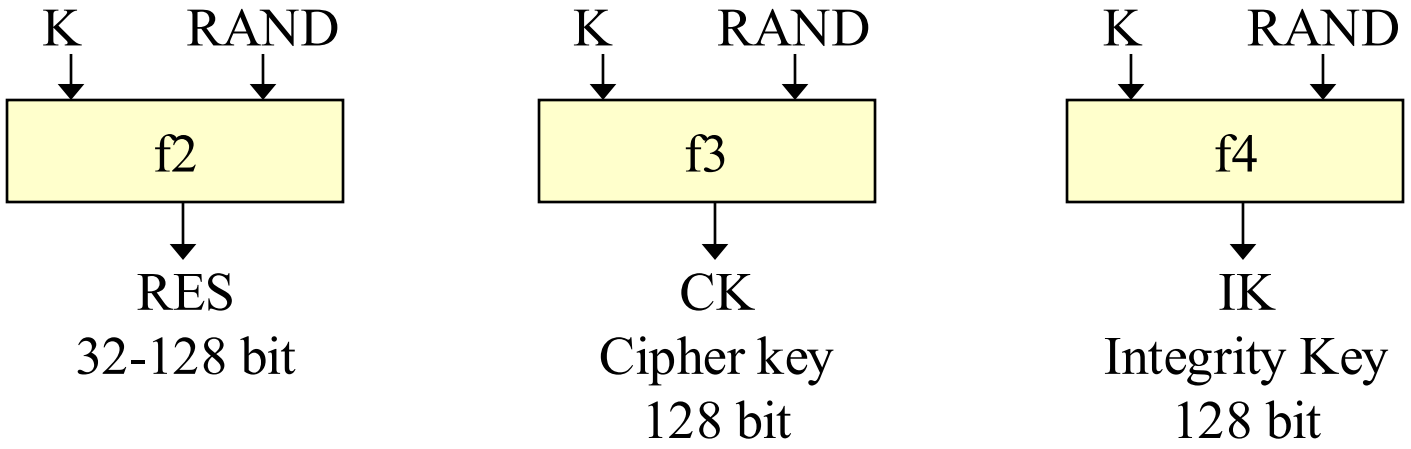


Authentication Vector:

- XRES (expected result)	as GSM SRES
- RAND	as GSM RAND
- AUTN (Netw. Auth. token)	- - -
- CK (cipher key)	as GSM Kc
- IK (integrity key)	- - -

MS authentication

➔**Usual challenge response, but different (public) algorithm**



Network Authentication

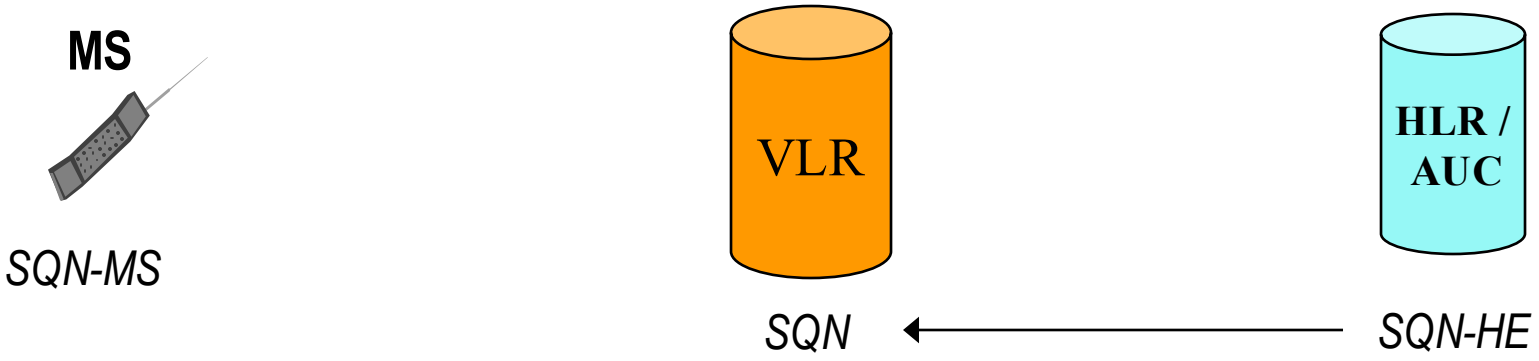
➔**MS should send a nonce…**

⇨*1 extra message*

➔**Bright idea: use Sequence number as "implicit" nonce!**

⇨*Issue: MS and AuC must be (approx) synchronized*

⇨*And robust procedures for resync must be specified*



SEQ as nonce: idea

Current SQN-MS
stored

Check
SQN = SQN-MS + 1
(or in appropriate
Tolerance range to
cope with lost msg)

SQN
(included in AUTN)
(given by AuC)

Use SQN as
“implicit” challenge!!

Once auth OK, update local SQN

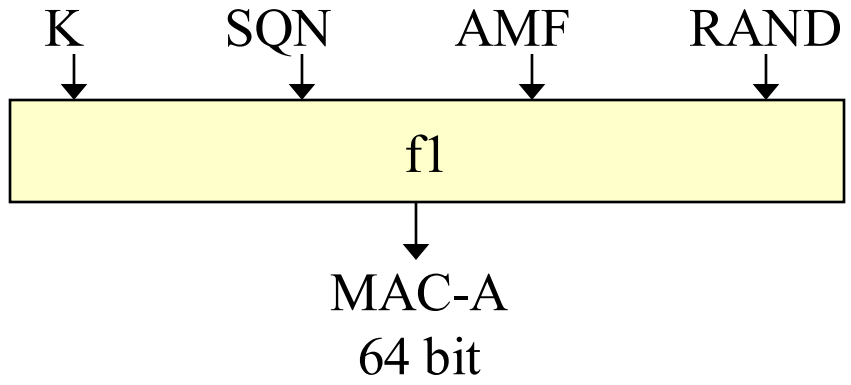
Network authentication: AUTN format

SQN *Sequence number*	*AMF* *Auth & key mgmt field*	*MAC-A* *Message Auth code*

48 bit

16 bit: carries info on which algo or key to use if choice available, sync window, etc (signalling info)

64 bit: allows MS to verify authenticity of Network!



MS: has all info needed to check that MAC-A transmitted by network is the same of MAC-A locally computed! SQN guarantees defense against replay

Minor detail: protecting SQN!

➔ **A privacy problem:**

⇨ *By looking at SQN (stepwise increasing), eavesdropper may discriminate and track user!*

➔ **Solution: mask SQN with Anonymity Key**

SQN xor AK	*AMF*	*MAC-A*
Sequence number	*Auth & key mgmt field*	*Message Auth code*

